

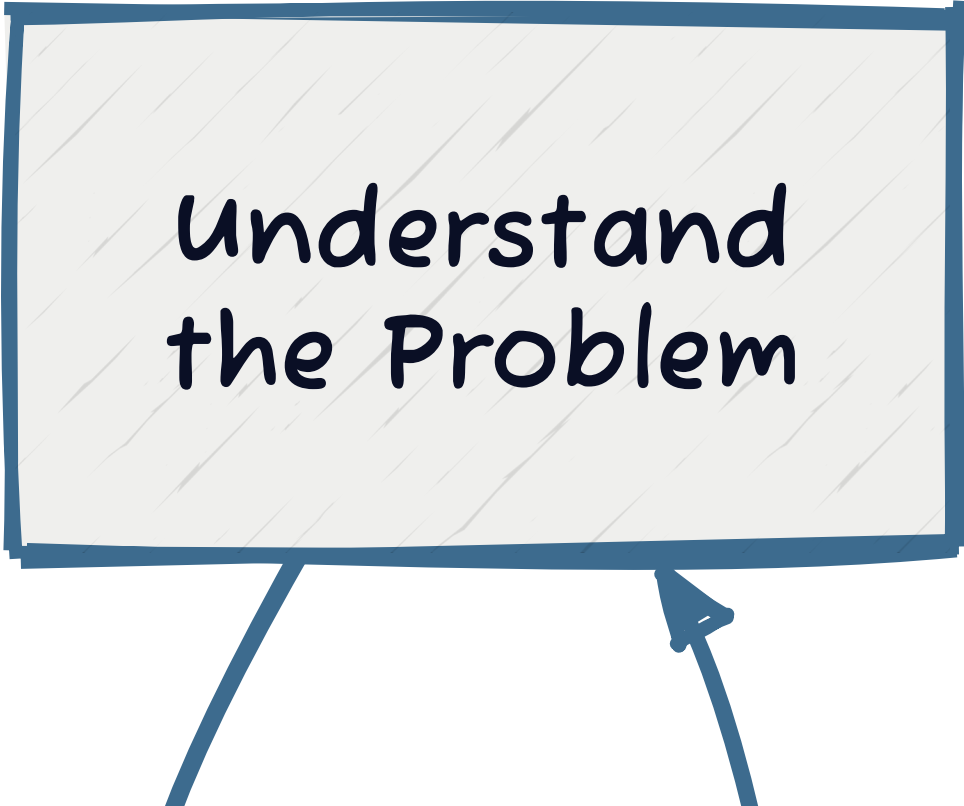
Day 04

The Modeling Project

MSU REU Machine Learning Short Course

The Full ML Loop

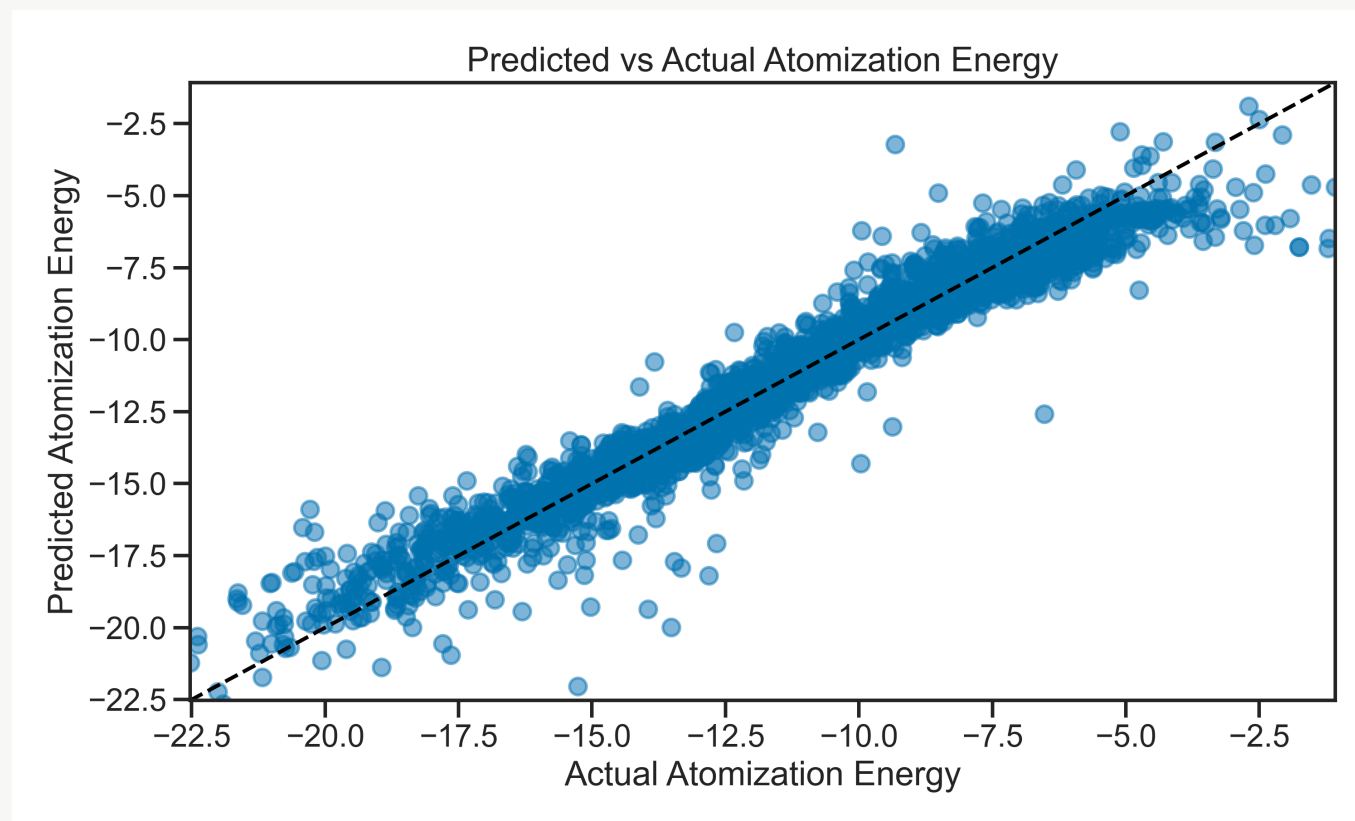
You've seen each piece. Today you run the whole thing yourself.



Understand
the Problem

Today's Dataset — Molecular Atomization Energy

Predict the energy required to pull a molecule apart, from its atomic structure.



1275 features from the Coulomb matrix representation of each molecule. Most of them are noise.

Start with a terrible model

Build the simplest possible model first.

```
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)

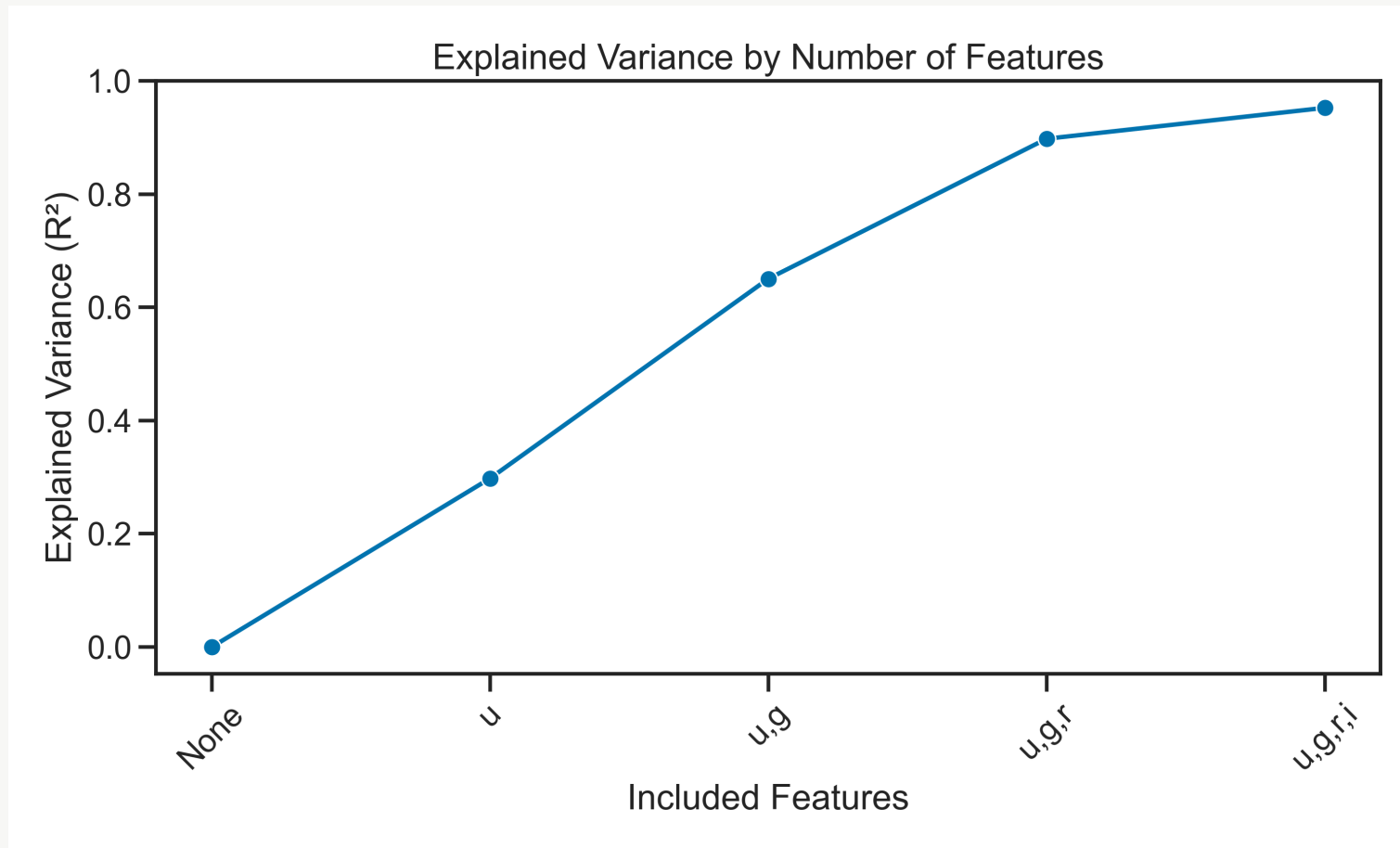
model = LinearRegression()
model.fit(X_train, y_train)
print(r2_score(y_test, model.predict(X_test)))    # probably bad
```

Then explore the data to understand **why** it's bad.

Hint: plot a histogram of each feature. What do you notice?

Feature selection — less is more

Not all 1275 features carry signal. Recursive Feature Elimination finds which ones to keep.



Cross-validation — how confident are you?

One train/test split gives you one number. Run the model many times to get a **distribution**.

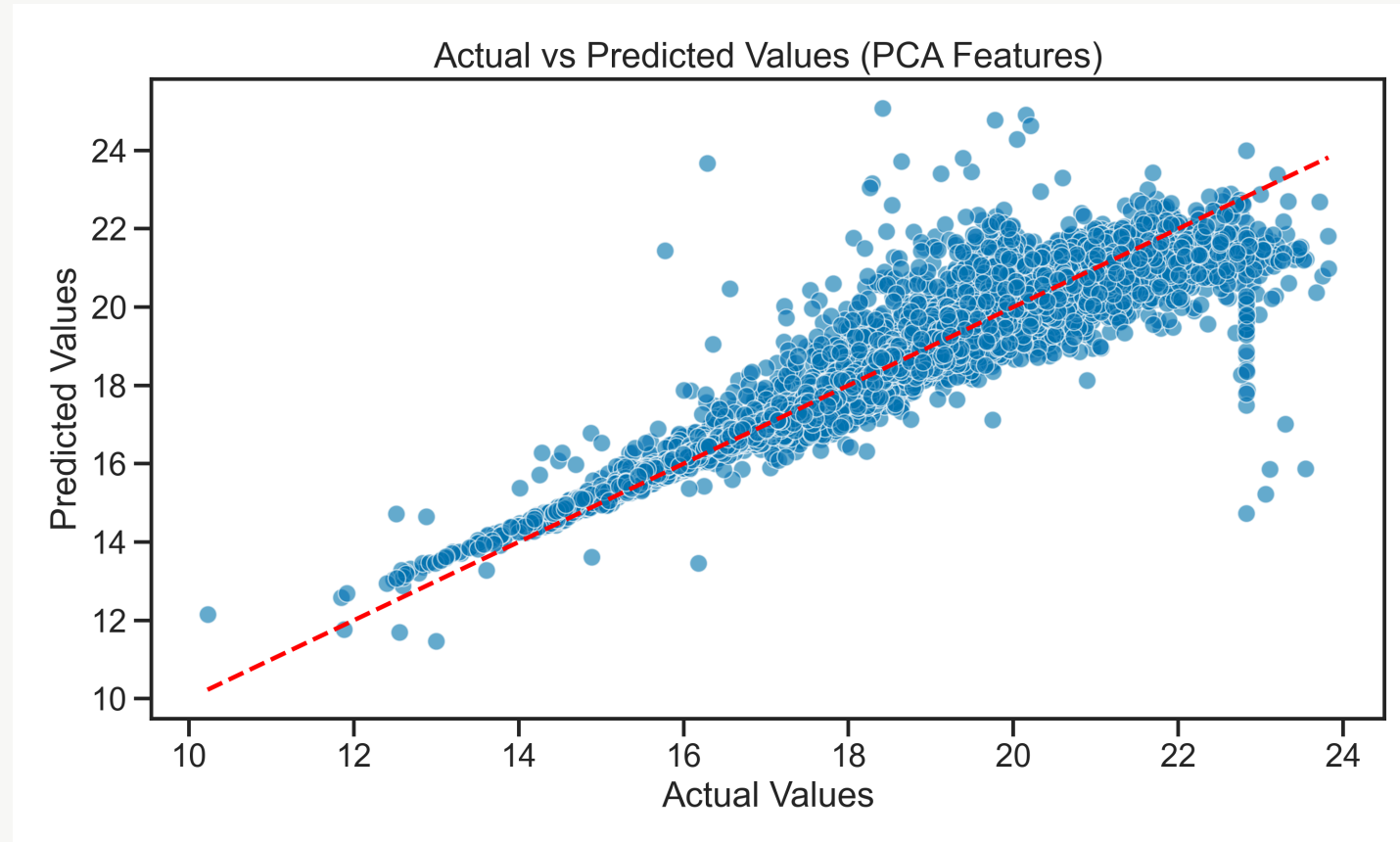
Each fold takes a turn as the test set.
This gives 5 independent estimates of model performance.

TEST	train	train	train	train
train	TEST	train	train	train
train	train	TEST	train	train
train	train	train	TEST	train
train	train	train	train	TEST

See: [Cross-Validation](#)

Dimensionality reduction — PCA

1275 correlated features → a small number of uncorrelated components that capture most of the variance.



Today's Activity

Work through **Activity 04: Modeling Project** — open-ended by design.

1. Load the molecular energy dataset
2. Build an initial model — expect it to be bad
3. Explore: why is it bad? Remove useless features.
4. Re-evaluate. Does it improve?
5. Implement cross-validation to estimate performance with uncertainty.

Notes: [Cross-Validation](#) · [RFE](#) · [PCA](#)

Use everything from Days 1–3. Ask questions. Improve iteratively.